Gianluca Di Paola

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M.Sc. student at Politecnico di Milano (GPA 3.83/4.0) and **first author of an ICSOC 2026 paper (CORE A)**, with experience in iOS, distributed systems, and LLM pipelines. Seeking software engineering and research roles.

EDUCATION

Politecnico di Milano 2025–Present

Master of Science Degree in Computer Engineering, ICT, Business and Innovation *Current GPA: 3.83/4.0 (28.69/30)*

Coursework: Advanced Software Engineering, Software Security, Advanced Database Systems, Advanced Computer Architectures, Performance Evaluation, Process and Service Design, Software Defined Networking, Algorithmic Game Theory

Honors: Merit-Based Tuition Fee Exemption awarded by Politecnico di Milano for particularly high academic merit

Politecnico di Milano 2021–2024

Bachelor of Science Degree in Engineering of Computing Systems

Coursework: Algorithms and Data Structures, Operating Systems, Software Engineering, Object-Oriented Programming, Computer Architecture, Databases, Human-Computer Interaction

PUBLICATIONS

G. Di Paola, A. Vitobello, E. Di Nitto, G. Quattrocchi. *Assessing the Robustness of LLM-Based Oracles for Blockchain-Based Services.* International Conference on Service-Oriented Computing (ICSOC 2026), Lodz, Poland, Dec 2026. CORE rank A. Accepted. DOI

- Evaluated the C-LLM oracle against coordinated malicious nodes (3 LLM families, 64 configurations): in 35, malicious nodes overtook honest ones in credibility; honest-response selection fell to 0% in the worst case.
- Compared with human-based oracles (Astraea, DeepThought). Artifacts: Zenodo, Software Heritage.

TECHNICAL SKILLS

Programming Languages: Python, Java, C, Swift, TypeScript, SQL, JavaScript, HTML, CSS, VHDL

Frameworks and Libraries: SwiftUI, JavaFX, React, Next.js, Tailwind CSS

AI and Research: LLM APIs (OpenAI, Gemini, DeepSeek), LLM agents, RAG, prompt engineering, LLM evaluation

Concepts: Object-oriented programming, design patterns, distributed systems, algorithms, software testing

Tools: Git/GitHub, Linux, Xcode, IntelliJ IDEA, VS Code, Maven, GCC, GDB, Valgrind, Vivado, Figma, LaTeX

Languages: Italian (native); English (C1 CEFR, Cambridge C1 Advanced, 2023)

PROJECTS

Skené | *Swift, SwiftUI, MVVM, Repository pattern, SQL, TMDB API, iCloud, Xcode* 2025–2026

- Team project with **Bending Spoons** at Politecnico di Milano (**30/30**): built and fully tested a native iOS and macOS app.

Virgin of the Rocks Reconstruction | *Imagen 3, Flux 1.1 Pro, Reve Image 1.0, prompt engineering* 2025

- Multidisciplinary Project, **30/30 cum laude** (advisor: Prof. Maristella Matera; with PhD student Luca Cordioli and the **Archivio di Stato di Milano**): transcribed and translated the late-15th-century commission contract, revealing a polyptych.
- Benchmarked three text-to-image models; chose Imagen 3 (zero-shot Leonardesque style) and rebuilt each polyptych panel.

Personal Portfolio | *TypeScript, Next.js, React, Tailwind CSS, shadcn/ui, Cloudflare Pages* 2025

- Solo open-source project: bilingual (EN/IT) static website with data-driven content for low maintenance.

VentureLabs Flâneur | *Figma, User-Centered Design, usability testing* 2024–2025

- Team project (HCI): needfinding, Figma prototypes, heuristic evaluation, and usability testing (SUS, SEQ).

AI Web Navigator | *Python, GPT, RAG, Newelle* 2024

- Team project, open source (55 GitHub stars): Newelle extension where an LLM agent browses web pages and cites sources.

Codex Naturalis | *Java, JavaFX, MVC, TCP, RMI, Maven* 2023

- Team of four (**30/30 cum laude**): distributed board game with GUI/TUI, live chat, multiple matches, and crash recovery.

Data Highway | *C, GCC, GDB, Valgrind, Unix Shell* 2022

- Solo project (**30/30**): memory-constrained route-planning engine in C for inputs up to 50 MB; validated on 111 test cases.

FPGA Module Design | *VHDL, Vivado, Artix-7 FPGA* 2022

- Team project: stream-processing module (10-state FSM, 52 flip-flops, 0 latches) with 16.2 ns slack at a 20 ns clock.

EXPERIENCE

Reply Student Tech Clash: AI Agents | *Finalist team, 2nd place*: built a GPT-based agent for polimi.it in 6 hours 2024

TOP: Tutoring Online Program | *Tutor*: tutored a student in difficulty; awarded an open badge 2023

Water and Data Analysis with Bioindicators | *Volunteer*: led a team water analysis of the Oreto river (Sicily Region) 2021

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